Research report

European banking customers data analysis

Completed on
Jun 25, 2026

Prepared by
Aditi Jain

Abstract

This study analyzes data of 10,000 customers from a European bank to understand why customers decide to leave. By performing data validation and cleaning , and analyzing segments like geography , age, bank balance, and account inactivity , some insights could be drawn. For instance , by analyzing factors uniquely and drawing interactions among demographic and financial factors, the results show that losing customers is usually caused by a mix of different factors working together. However, a few groups stood out as being at a higher risk , specifically customers belonging in Germany and between 46 to 60 years old and those who aren't very active with their accounts. Based on these key insights, some business implications and recommendations have been put up to help financial institutions. At last, while this study provides a solid foundation, it also highlights the future steps to be taken such as using these patterns to build models that predict who might leave before they actually do.

Index..........2

1. Introduction

Customer retention has become a critical priority for financial institutions . Customer churn, defined as the loss of customers who discontinue their relationship with a bank, can significantly impact profitability, as it represents one of the largest hidden costs in retail banking. Losing existing customers leads to reduced lifetime value , increased acquisition costs and revenue instability. Hence this project aims to identify critical insights by conducting data analysis of European banking dataset.

Problem Statement

• Identifying high-risk customer segments

• Understanding churn differences by geography and demographics

• Quantifying the financial profile of churned customers

Project Objectives

• Measure overall churn rate

• Identify churn distribution across customer segments

• Compare churn behavior across European regions

Secondary Objectives

• Understand churn among high-value customers

• Evaluate engagement and tenure patterns

• Support strategic planning and marketing decisions

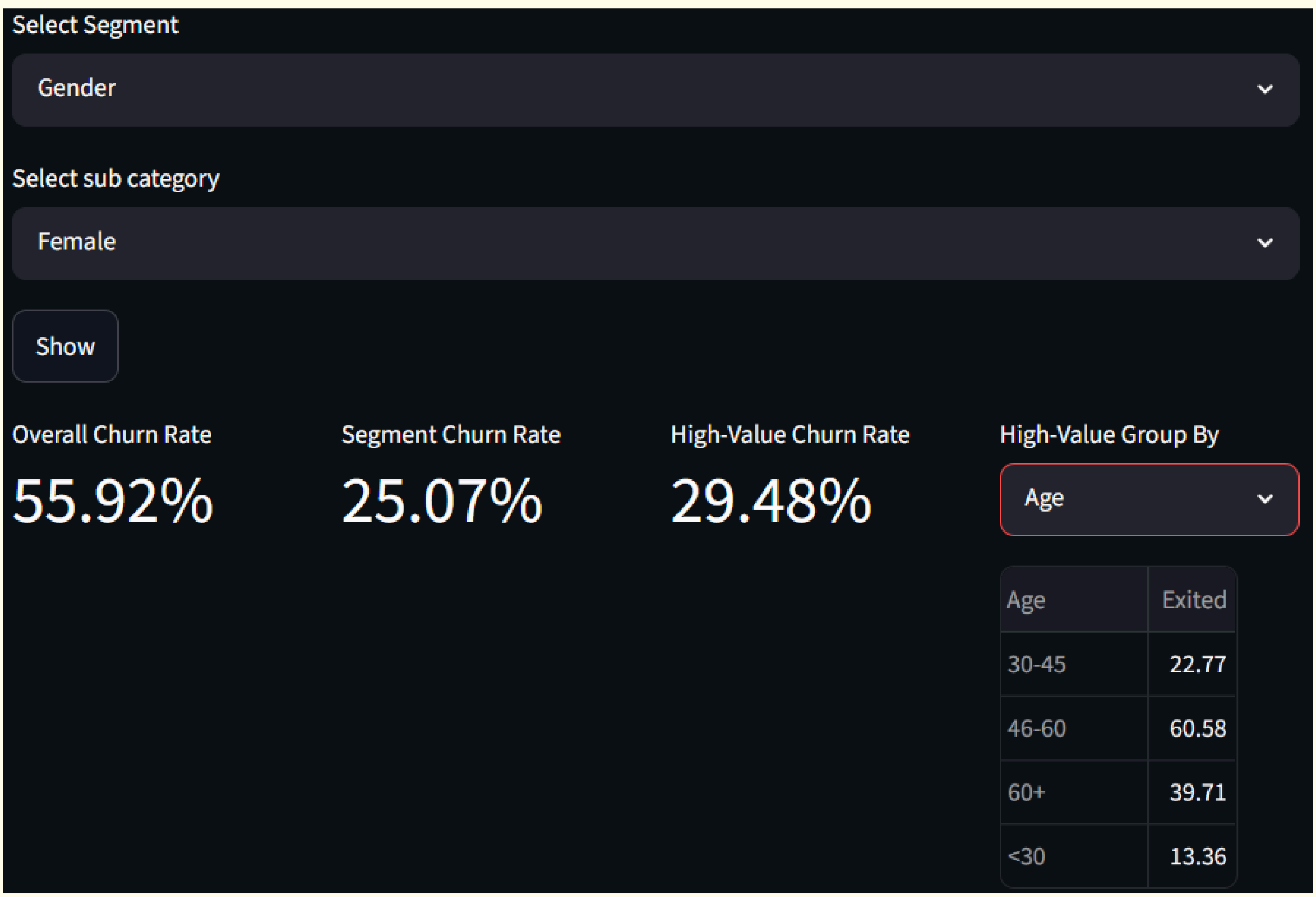
2. Dataset Description

The dataset contains **10,000 bank customers** with demographic, financial, and account information. The target variable, **Exited**, indicates whether a customer left the bank (1) or remained (0).

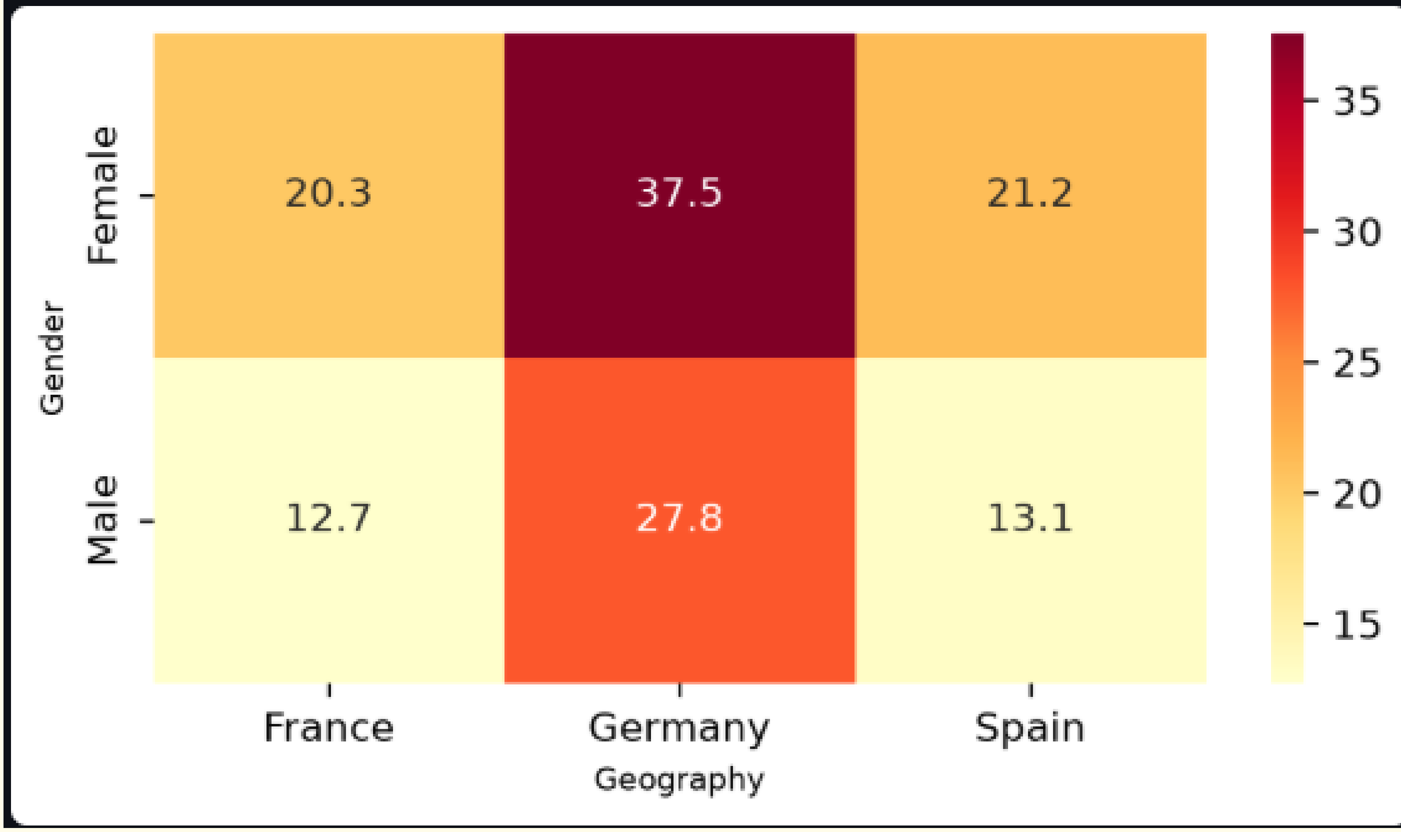
Column	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer creditworthiness
Geography	France, Spain, Germany
Gender	Male / Female
Age	Customer age
Tenure	Years with the bank
Balance	Account balance
NumOfProducts	Number of bank products
HasCrCard	Credit card ownership
IsActiveMember	Activity indicator
EstimatedSalary	Estimated annual salary
Exited	Churn indicator **(target)**

3. Key Performance Indicators (KPI)

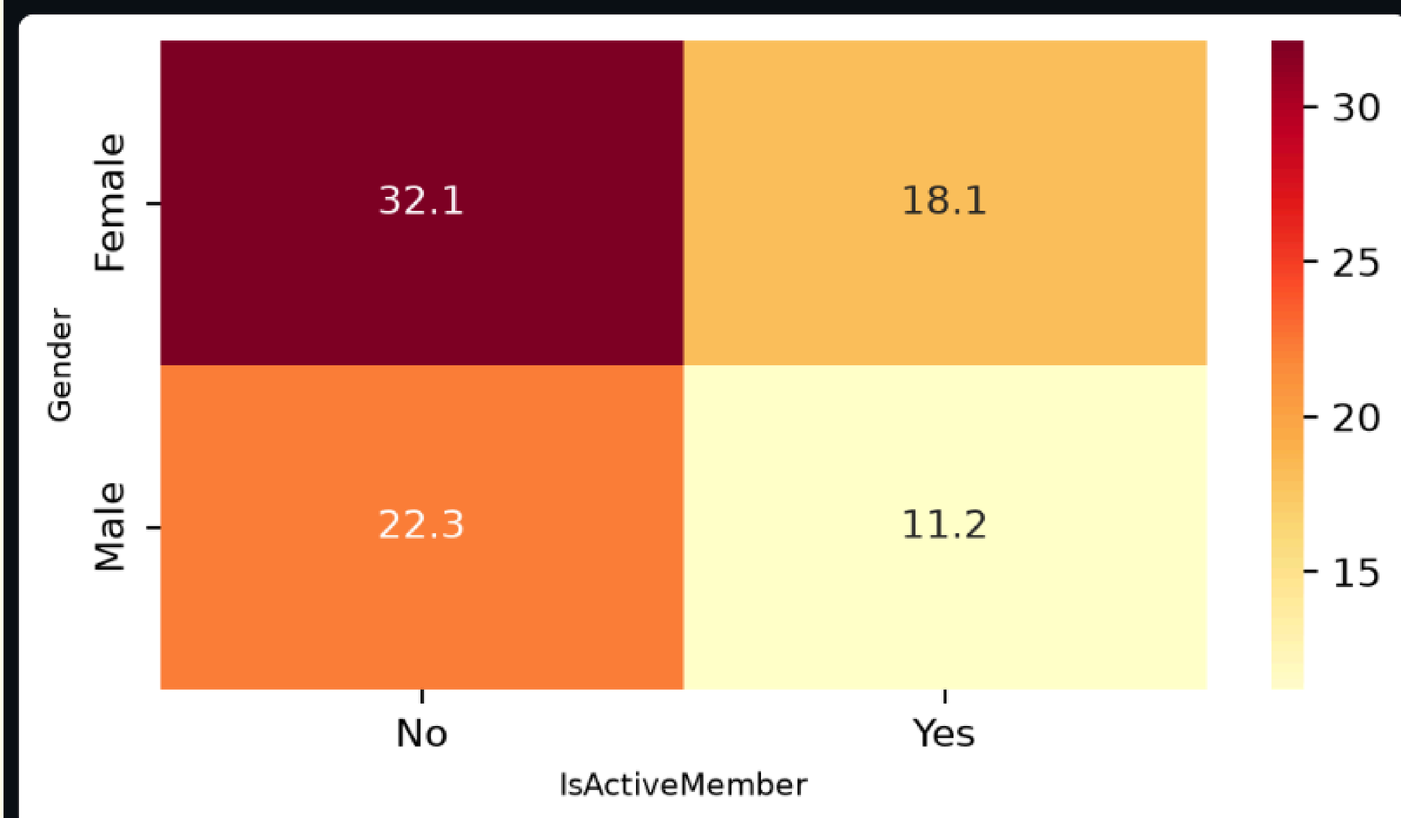
KPI Name	Description
Overall Churn Rate	% customers who exited
Segment Churn Rate	Churn % by segment
High-Value Churn Ratio	Churn among premium customers
Geographic Risk Index	Regional churn exposure
Engagement Drop Indicator	Inactivity vs churn



Geographic Risk Index



Engagement Drop Indicator Churn Rate(%)



4. Methodology

- ## Data Ingestion And Validation
  - Load dataset
  - Validate engagement and product fields
  - Ensure binary variables consistency
  - Confirm churn labeling accuracy

- ## Data Cleaning & Preparation
  - Remove duplicate records
  - Remove non-analytical fields (surname , year , customer_id)
  - Check spelling errors in categorical columns

- ## Customer Segmentation Design
  - Check Outliers
  - Geographic Segmentation: France, Spain, Germany
  - Age Segmentation: <30, 30–45, 46–60, 60+
  - Credit Score Bands: Low, Medium, High
  - Tenure Groups: New, Mid-term, Long-term customers
  - Balance Segments: Zero-balance, Low-balance, High-balance

- ## Churn Distribution Analysis
  - Overall churn rate
  - Segment-wise churn rates
  - Churn contribution by segment size
  - Comparison of churned vs retained profiles

- ## Comparative Demographic Analysis
  - Gender-based churn differences
  - Geography-age interaction analysis
  - Financial stability vs churn comparison

- ## High-Value Customer Churn Analysis
  - Identify high-balance churners
  - Compare salary vs balance churn patterns
  - Quantify revenue risk from churn

5. Exploratory Data Analysis

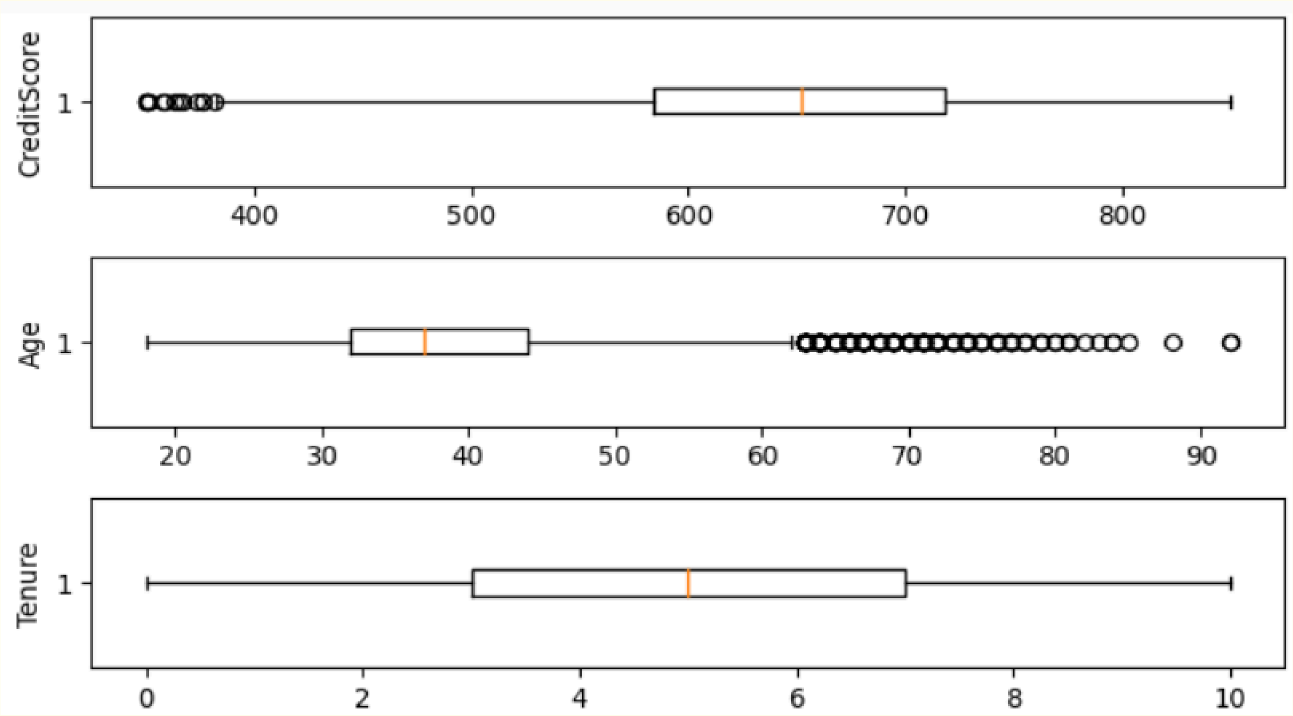
5.1 Data Ingestion And Validation

Check	Result
Data types	Valid
Missing values	0
Binary values were present in columns HasCrCard, IsActiveMember, and Exited	All Valid (0,1)
Negative values	None

5.2 Data Cleaning & Preparation

Check	Result
Duplicate records	0
Spelling Mistakes in categorical columns	None

5.3 Customer Segmentation Design



CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
Medium	France	Female	30-45	New	Zero_Bal	One	Yes	Yes	Mid	1
Medium	Spain	Female	30-45	New	Low_Bal	One	No	Yes	Mid	0
Low	France	Female	30-45	Long-Term	High_Bal	Multiple	Yes	No	Mid	1
Medium	France	Female	30-45	New	Zero_Bal	Multiple	No	No	Mid	0
High	Spain	Female	30-45	New	High_Bal	One	Yes	Yes	Mid	0
...	...	...	...	...	...	...	...	...	...	...
High	France	Male	30-45	Mid-Term	Zero_Bal	Multiple	Yes	No	Mid	0
			30-	Long-						

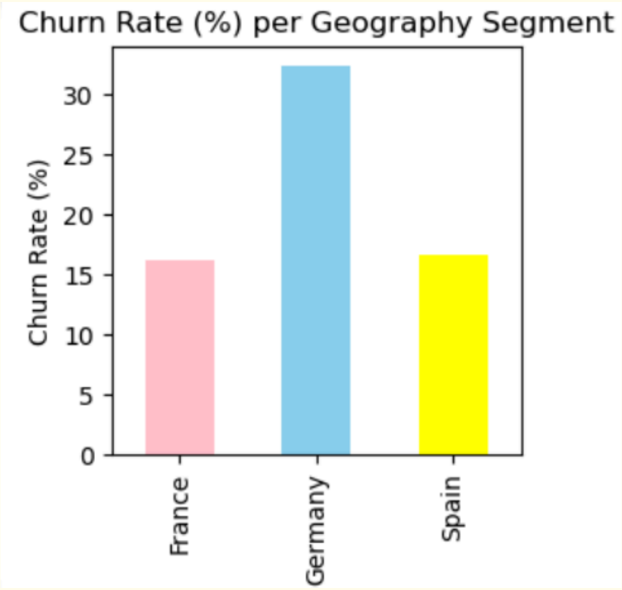
Check	Result
Outliers	Age and credit score being 60+ and below 400 respectively is possibly valid.

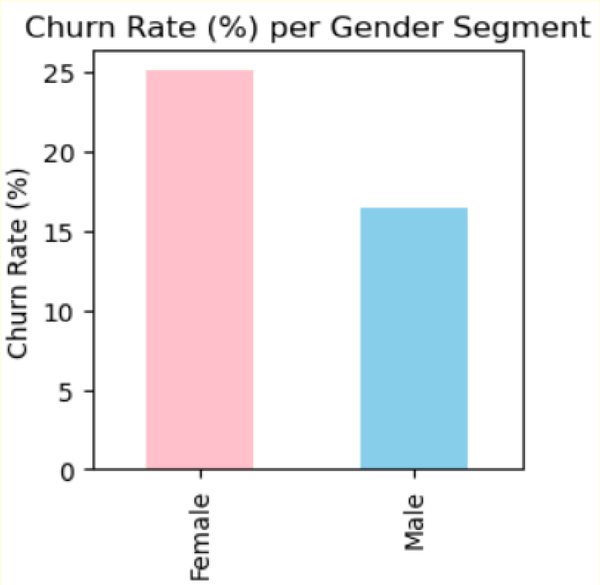
5.4 Overall Churn Distribution

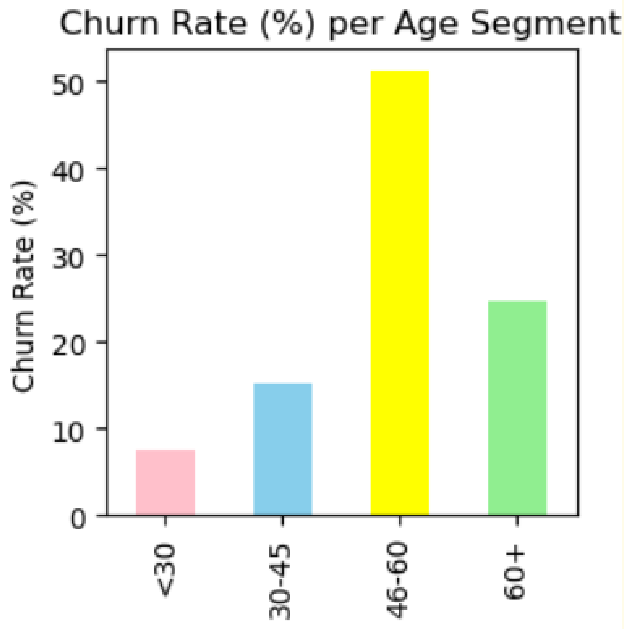
Total Customers	Churned Customers	Churn Rate	Retained Customers	Retention Rate
10000	2037	20.37%	7963	79.63%

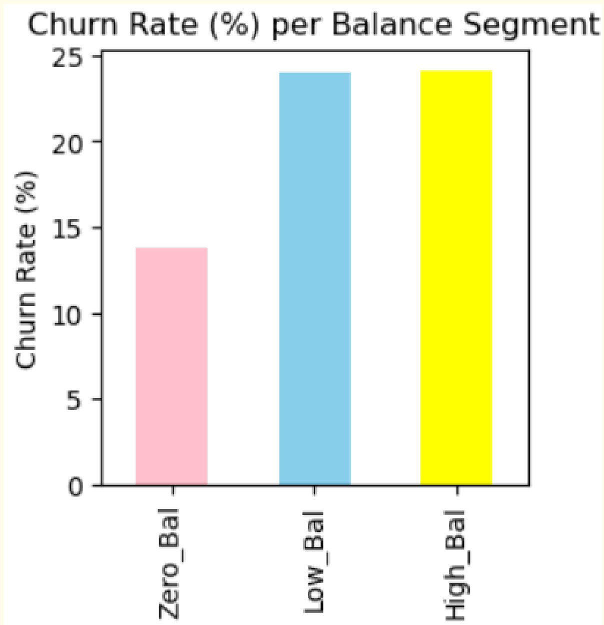
	Segment	Highest Churn Rate	Highest Churn Contributor	Highest Retain Rate Per Segment	Highest Retained Contributor
0	CreditScore	Low	Medium	Medium	High
1	Geography	Germany	Germany	France	Spain
2	Gender	Female	Female	Male	Male
3	Age	46-60	30-45	<30	60+
4	Tenure	New	New	Long-Term	Mid-Term
5	Balance	High_Bal	High_Bal	Zero_Bal	Zero_Bal
6	NumOfProducts	One	One	Multiple	Multiple
7	HasCrCard	No	Yes	Yes	No
8	IsActiveMember	No	No	Yes	Yes
9	EstimatedSalary	High	Mid	Mid	Low

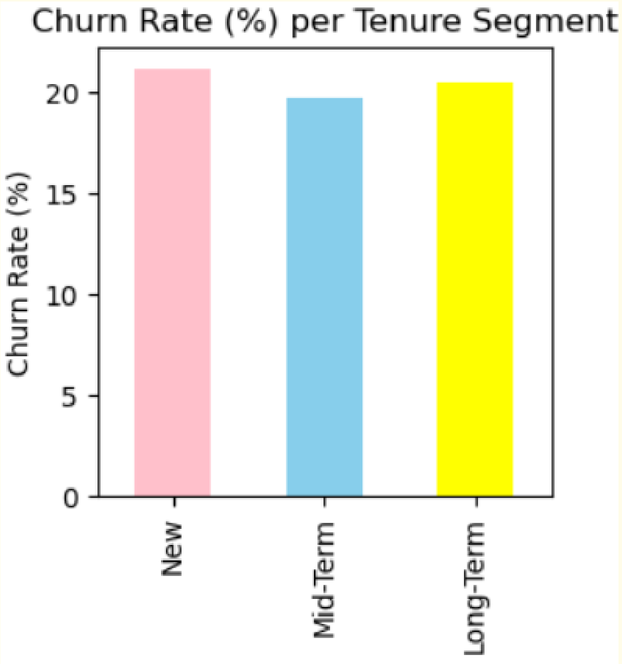
5.5 Univariate Analysis - Segment-wise churn rates

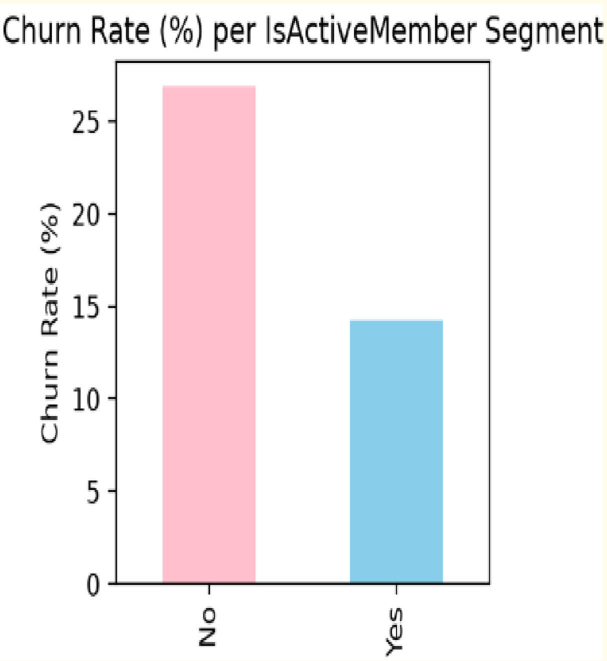




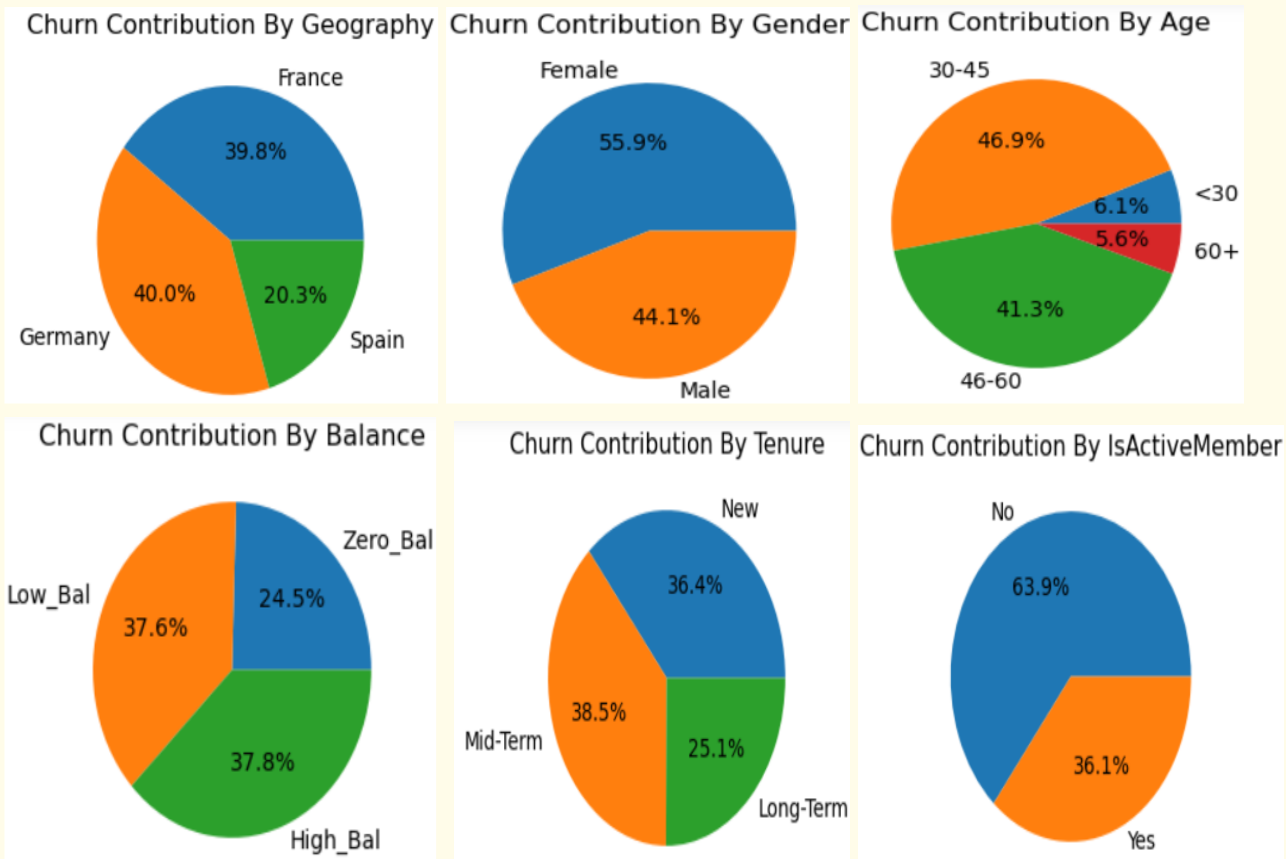








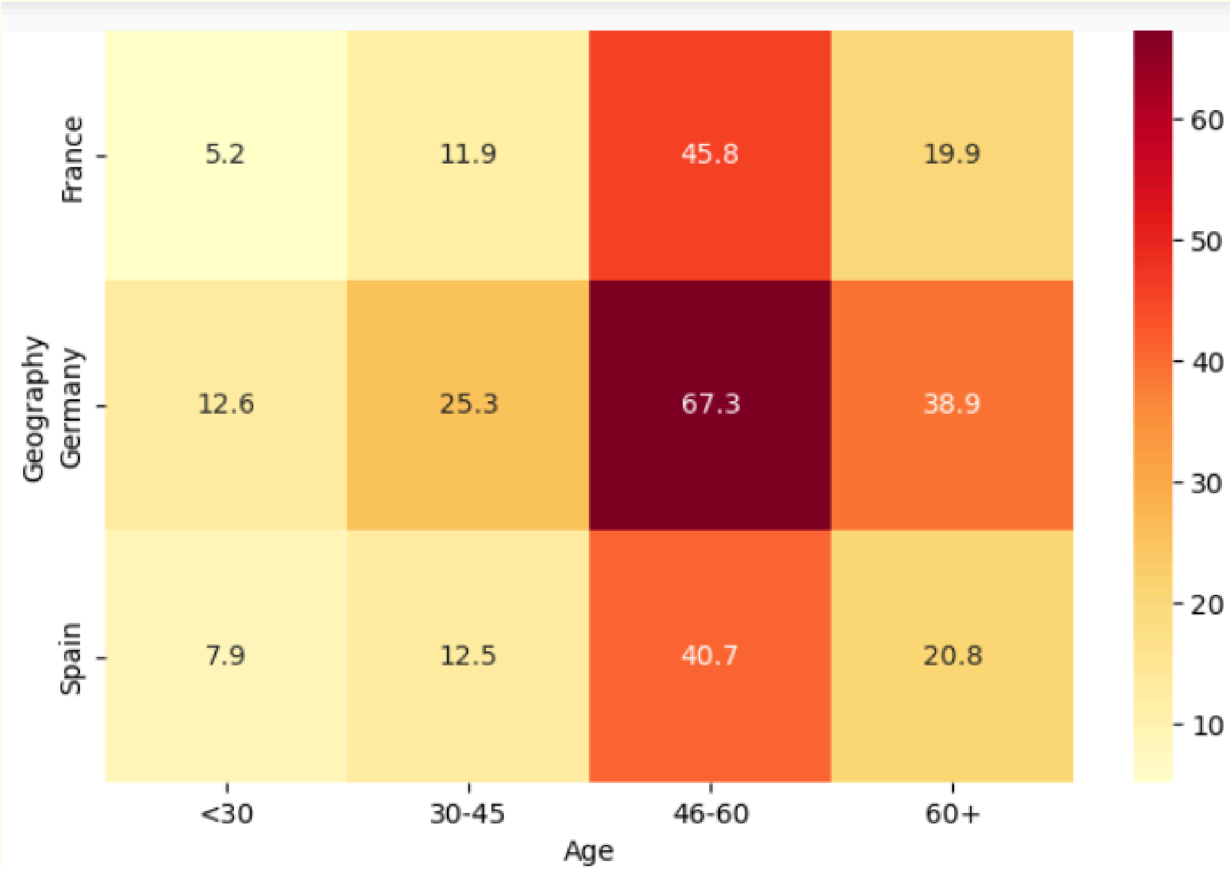
5.6 Univariate Analysis - Churn contribution by segment size



Parameter	Insight
Segment wise churn rate	Answers "Which segment is most likely to leave?" For ex. Geography-wise - **35% Germany** customers are likely to churn compared to **15% French.** This metric identifies **high-risk segments.**
Churn contribution per segment	Answers "Which segment contributes the most churned customers to the business?" For ex. **40%** churned customers belong to **France, same as Germany** , even though its churn rate is **lower** than that of Germany. This metric identifies **where the business impact is largest.**

5.7 Demographic Analysis

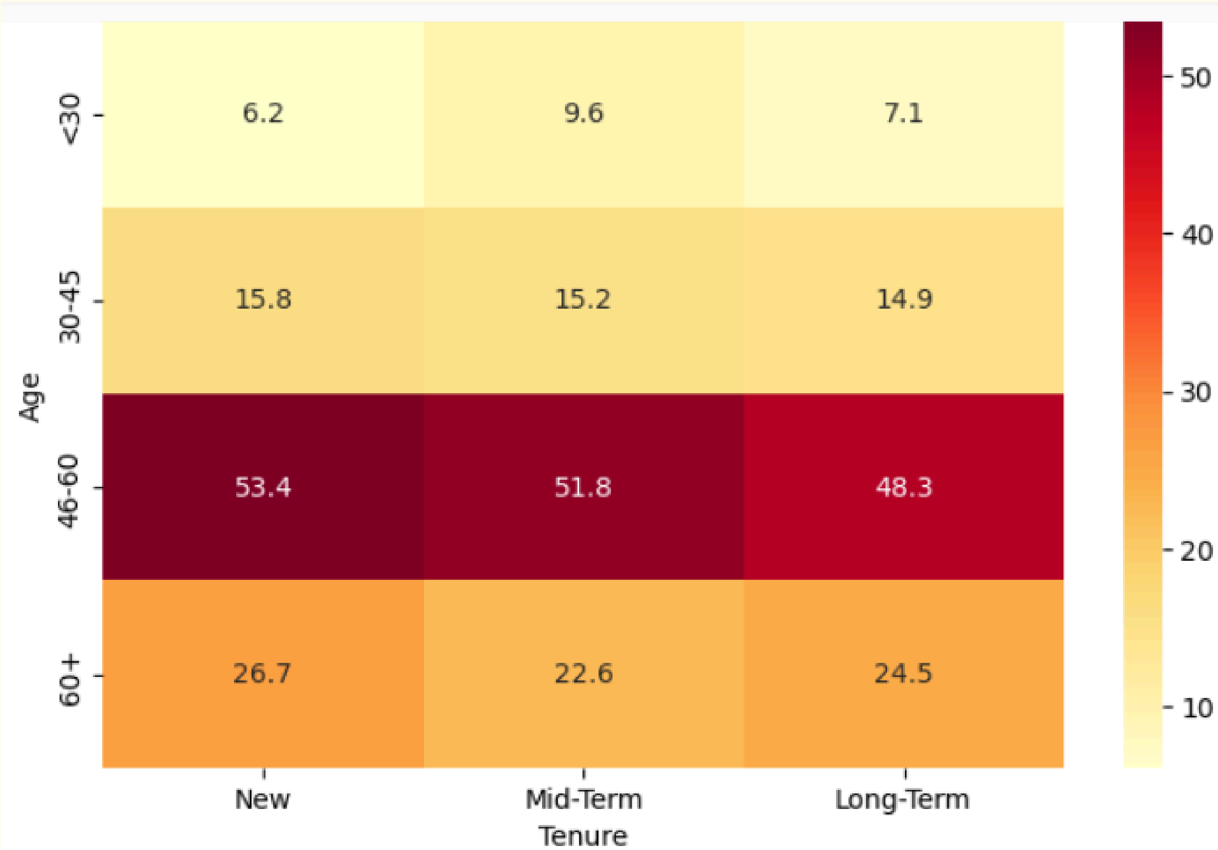
(a) Geography - Age Interaction



Insight

Customers aged **46–60** in **Germany** show the highest churn rate, suggesting regional and age-related factors jointly influence churn.

(b) Age - Tenure Interaction

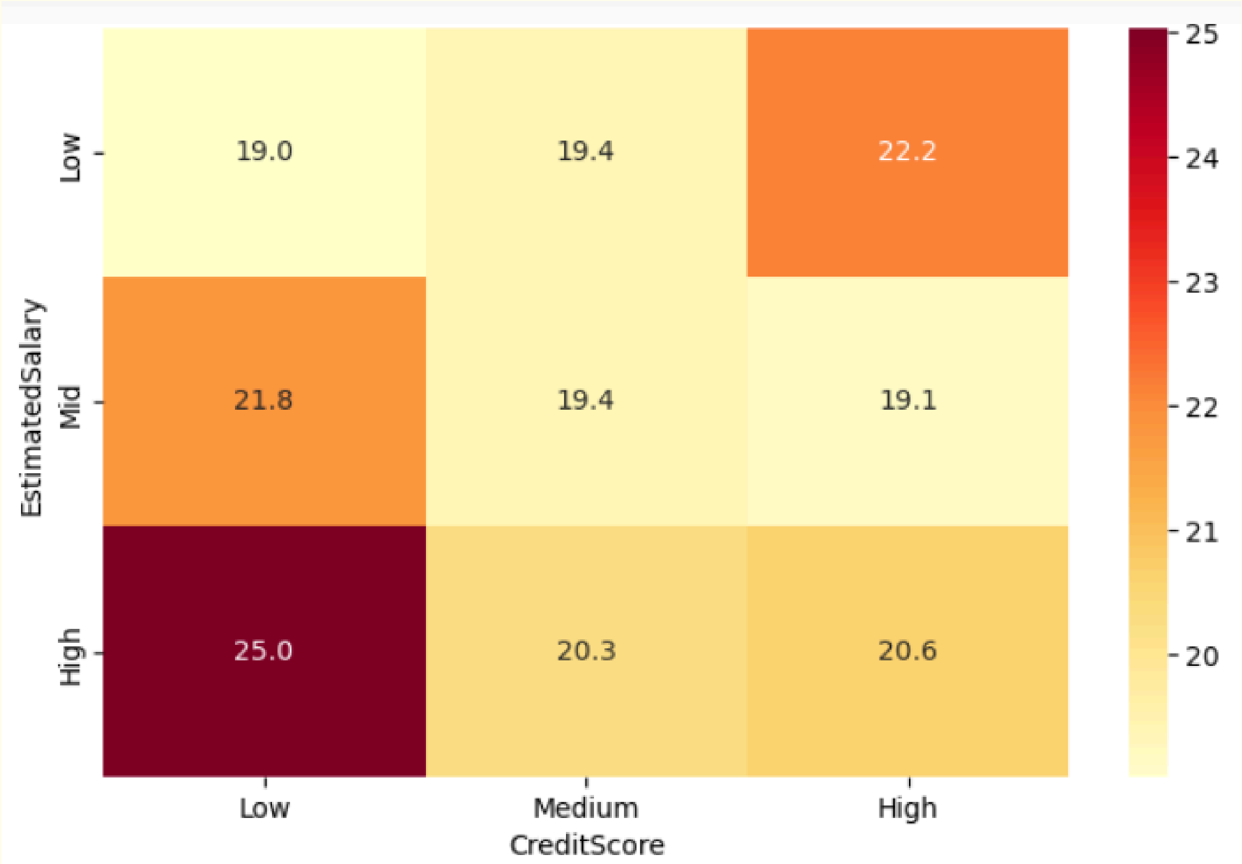


Insight

Customers aged **46–60** who are **newly** associated with bank for **less than 3 years** show the highest churn rate, however it is **closely similar for mid tenure** customers as well (3-7 years). Hence churn is **equally spread** across tenure when compared within each age group.

5.8 Financial Analysis

(a) Estimated Salary Vs Credit Score



Insight

Low-income customers with poor credit scores tend to stay as due to their credit history , they may have fewer alternative banking options in comparison to high salaried. Showing that the latter can easily switch to competitors offering better products or loan terms. Hence , **high-income customers with poor credit scores are a high-risk churn segment.**

5.9 High Value Customer Churn Analysis

Parameter	Value
Total number of High Value Customers	3191
Total number of High Value churners	771
Churn rate among High Value customers	24.16
Retention rate among High Value customers	75.84

	Segment	Highest Churn Rate	Highest Churn Contributor	Highest Retain Rate Per Segment	Highest Retained Contributor
0	CreditScore	Low	Medium	High	High
1	Geography	Germany	Germany	Spain	Spain
2	Gender	Female	Female	Male	Male
3	Age	46-60	30-45	<30	60+
4	Tenure	Mid-Term	Long-Term	New	Mid-Term
5	NumOfProducts	One	One	Multiple	Multiple
6	HasCrCard	No	Yes	Yes	No
7	IsActiveMember	No	No	Yes	Yes
8	EstimatedSalary	Low	Mid	Mid	High

6. Key Insights

- The overall customer churn rate is **20.37%**, indicating that approximately **one in every five customers** has discontinued their relationship with the bank.

- Demographically, **Germany and France contribute a similar share of total churn**, while customers aged **30–45** and **46–60** account for the largest proportion of churned customers. However, **female customers aged 46–60 residing in Germany exhibit the highest churn rate**, making them the highest-risk demographic segment.

- Inactive customers are **approximately 10 percentage points more likely to churn** than active customers, highlighting customer engagement as a key retention factor. Additionally, customers holding only **one banking product** exhibit a higher churn tendency than those using multiple products.

- Financially, customers with **mid-to-high incomes and high account balances** contribute the largest share of churn and also exhibit relatively high churn rates. In contrast, customers with **zero account balances** demonstrate comparatively high retention.

- Among high-value customers, the churn rate is **24%**, indicating that **approximately one in every four** high-value customers is at risk of leaving the bank.

- Interaction analysis indicates that customers with **high account balances, low incomes, and high credit scores** are more likely to churn, whereas customers with **mid incomes** within the same balance segment exhibit comparatively lower churn.

- Customer attrition represents an estimated **€18 million** in lost account balances, highlighting the significant financial impact of churn on the bank.

7. Recommendations

a) Based on the findings, following actions should be taken by bank to reduce segment wise customer churn :

Insight	Recommendation
Germany has highest churn	Conduct customer surveys , engage in feedback calls and improve **localized banking services**. Focus on both customer and employee side feedback.
Age 46–60 churns the most	Introduce personalized relationship managers and loyalty programs for middle-aged customers.
Inactive members churn more	**Regular engagement** with product - through product email, SMS, exclusive user webinars , and mobile app notifications to encourage regular account activity.
New tenure customers show high churn	Strengthen onboarding with interactive tutorials, personalized guidance, and hands-on support for **clear on-boarding process** and to have strong start with banking product/service during the first three years.
Females churn more than men in every region	Move away from skewed 'male default' design and offer more personalized experience and engaging UX . Introduce **woman life tailored-product offering** during their life milestones (e.g., career breaks, education, maternity, and family planning)
High Value Customers	Assign dedicated **relationship managers** to reduce the financial impact of churn, provide them with exclusive benefits, priority customer support, and personalized financial advice.
High income customers and poor credit score have high churn	Offer personalized loan restructuring, **credit improvement programs**, and premium financial advisory services to retain these customers.
High retention customers	Focus on relationship marketing , develop **exclusive loyalty programs** and benefits.

b) Systematic recommendations:-

- Act before churn - Develop a machine learning–based churn **prediction model** that integrates demographic, financial, and behavioral variables to identify customers who are prone to churn .

- Implement a churn monitoring dashboard that **tracks churn rates** across key customer segments in real time.

- Pay attention to complaints - review customer engagement metrics to detect early signs of dissatisfaction as Zendesk reports that **56% of consumers rarely complain** and quietly switch to competitors.

- Hence , make **friendly ui/ux for providing feedback , categorise complaints** by identifying the stage and type of service (ex.product, onboarding, support, expectation gap)

8. Conclusion

Thus, this project provides a clear, segmentation-driven understanding of customer churn in European banking.

By uncovering churn patterns across geography, demographics, and financial profiles, it equips decision-makers with effective insights obtained through EDA. And based on these insights , it is recommended that banks adopt targeted retention strategies, strengthen customer engagement, focus on retaining high-value customers, and use data-driven approaches such as machine learning models to identify customers at risk of churn. This can improve customer satisfaction, reduce revenue loss, and enhance long-term profitability.

9. References

1. https://projects.unifiedmentor.com/project_instructions?id=13937
2. https://www.tigeranalytics.com/perspectives/blog/reducing-customer-churn-how-the-banking-sector-can-thrive/
3. https://www.unblu.com/en/blog/how-to-reduce-customer-churn-in-banking
4. https://www.superoffice.com/blog/reduce-customer-churn/
5. https://www.outreach.ai/resources/blog/reduce-customer-churn
6. EDA GEEKS